Utilization of Sustainable Infrastructure for Developing Countries

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Abstract. This paper asks how developing countries can reliably expand essential infrastructure while keeping environmental consequences in mind. It reviews World Bank research, peer-reviewed urban sustainability work, and institutional cost analyses. Infrastructure is a vital accelerant of economic and social progress: a meta-analysis of 221 studies finds a positive and significant impact of infrastructure on development outcomes. Its environmental costs are equally documented. During the twentieth century roadkill surpassed hunting as the leading direct cause of vertebrate mortality on land, and while pavement covers less than 1 percent of United States land area, the road-effect zone reaches up to 20 percent, with roughly 25 million kilometres of new road projected by 2050. Sustainable infrastructure is presented as the alternative. Nature-based infrastructure in cities is on average 42 percent cheaper than grey infrastructure delivering the same services and generates 36 percent more value for society, though the cost figure holds only when the wider benefits of nature-based systems are counted rather than construction price alone. Sustainable infrastructure outperforms conventional infrastructure by over 20 percent under a net-zero scenario and by roughly 10 percent under limited climate action, against a global infrastructure investment gap of approximately $15 trillion by 2040. Two limitations are acknowledged: the cost research concentrates on particular regions, and sustainable development does not resolve the planning and corruption problems that weak governments face.

Keywords. sustainable infrastructure; developing countries; nature-based solutions; road ecology; infrastructure finance

1. INTRODUCTION

Infrastructure is a vital part of our modern society, providing the backbone for many services taken for granted. The earliest recorded example of infrastructure that historians have found is the Natchez Trace, a trail devised by Native Americans 2000 years ago, running from Tennessee to Natchez, Mississippi, in order to follow large herds of bison during migration (Reliance Foundry, n.d.). Electricity, public transportation, and access to public healthcare and education are all common examples of the impact that infrastructure has in our everyday lives, and it would be close to impossible to function without them.

However, even though these types of infrastructure are a vital part of day-to-day life, they have their own adverse effects, especially on our natural world. According to an article written by Ben Goldfarb, an environmental journalist, during the twentieth century roadkilling surpassed hunting as the leading direct cause of vertebrate mortality on land, and roads kill more creatures with less fanfare than any other human activity (Goldfarb, 2023).

For developing countries, this conflict is especially important. They often face a great deal of pressure to build essential infrastructure while still needing to keep environmental impacts in mind. This raises the question: how can developing countries reliably expand essential infrastructure while still keeping environmental consequences in mind? In this paper, it will be shown how sustainable infrastructure is the most reliable method for developing countries to meet their developmental needs while still keeping the environment as a major focus.

2. IMPORTANCE OF INFRASTRUCTURE DEVELOPMENT IN DEVELOPING COUNTRIES

Infrastructure is a vital accelerant of not only economic but also social progress in developing countries. In terms of the economy, infrastructure is vital in boosting productivity and connectivity, and it allows products and labor to move around efficiently. According to Timilsina et al. (2023), investing in physical infrastructure such as transportation can help reduce overall trade costs and increase economic output. For example, the improvement of trade systems can help connect producers and consumers, which furthers both economic and national development, because markets in these countries are usually broken apart and can be hard to access.

In addition to physical infrastructure, non-physical infrastructure such as education is also very important in the development of countries. An analysis of 221 studies conducted by Vivien Foster, an economist with the World Bank, and colleagues found a positive and significant impact of infrastructure on development outcomes (Foster et al., 2025). Foster et al. write in coordination

with the World Bank, an international finance institution, which raises their credibility while also lowering it, as they may exclude the limitations of such infrastructure development. However, due to the sheer amount of evidence analyzed, the overall trend shows that infrastructure has a positive impact on development outcomes, supporting the idea that infrastructure is the foundation of development. All of these investments combined help show how infrastructure can contribute to long-term growth.

Infrastructure does not only contribute to the economy, it can also help sustain long-term social development. According to research conducted by Jennifer Ferreira, a researcher based in the Coventry University Centre for Business, increased access to shared public spaces can help reduce social isolation and improve overall wellbeing (Ferreira et al., 2021). This is significant in developing countries because the main focus is usually on other forms of infrastructure, such as energy systems or communication networks, and these shared spaces are overlooked. In addition, the rapid urbanization that occurs in these countries can weaken the impact of traditional social structures. By strengthening these social connections and inclusivity, infrastructure can contribute not only to economic growth but also to stronger and more cohesive societies.

3. ENVIRONMENTAL IMPACTS OF INFRASTRUCTURE DEVELOPMENT

In spite of the positive effects of infrastructure, its development can still harm the environment, especially when sustainability is not one of the major focuses. One major issue is that infrastructure development disrupts ecosystems. Major projects usually require a great deal of land clearing, which destroys natural habitats, alters ecological systems, and leads to a loss of biodiversity.

According to van Oorschot et al. (2024), a group of urban sustainability researchers, replacing green infrastructure, which they define as a strategically planned network of natural and semi-natural areas designed and managed to deliver a wide range of ecosystem services, with grey infrastructure, which they define as traditional infrastructure, can increase emissions and make environments less resilient over time. This can lead to further destabilization of the environment, which can cause long-term consequences.

Furthermore, infrastructure plays a critical role in facilitating an increased amount of pollution. According to the World Bank (n.d.), the construction processes and increased industrial activity that are commonplace with infrastructure development can cause soil, air, and water contamination. When combining this factor with the others, the destruction of natural habitats, the alteration of ecosystems, and the loss of biodiversity, environmental damage builds up over time, making environments less resilient and leading to potential collapse. This highlights a potential flaw in infrastructure planning: prioritizing economic development and short-term gains over the wellbeing of the environment and the long-lasting impacts it holds for the future.

This type of infrastructure investment can also majorly affect biodiversity. According to Goldfarb (2023), studies have found that while actual pavement makes up less than 1 percent of the United States, its influence, the road-effect zone, covers up to 20 percent of the country's land area. In developing countries, where road expansion is still pushing into intact habitats, the ecological impact is likely to be worse than those figures suggest, though no equivalent measurement for those settings is available.

Research conducted by the International Fund for Animal Welfare (2025), an international organization dedicated to the wellbeing of animals, contributes to this point by explaining how breaking up habitats can isolate certain animal species, which harms their ability to reproduce. The impact of this does not just threaten individual species, but entire ecosystems. When putting all of these impacts together, it shows that while infrastructure is necessary for development, it can still cause serious risks for the natural world.

4. WHY IS SUSTAINABLE INFRASTRUCTURE A VIABLE REPLACEMENT?

Now that the environmental impacts of infrastructure have been discussed, a potential alternative should be explored, specifically sustainable infrastructure. Goldfarb's own research points toward its necessity. According to Goldfarb (2023), road ecology has revealed the perils of reckless development and pointed us toward solutions. This is especially important for developing countries, as many have not yet laid down their central infrastructure, which gives them the opportunity to get it right from the beginning rather than coming back and trying to fix it decades later.

According to research conducted by Christian Deseglise and Louis Downing, sustainable infrastructure outperforms conventional infrastructure by over 20 percent under a net-zero scenario, with cumulative returns roughly 10 percent higher under limited climate action (Deseglise & Downing, 2025). The margin therefore depends on which climate path the world follows, but it favours the sustainable option under both. This shows how sustainable infrastructure can benefit developing countries by giving them an economic advantage while they remain under economic constraints.

In addition, sustainable infrastructure is helpful for developing countries because it can help them overcome financial constraints. According to Fay et al. (2017), most developing countries have severe financial constraints due to the limited amount of money they have and the number of things that need to be repaired or built in the first place. According to research done by Ronja Bechauf, a policy advisor at the International Institute for Sustainable Development, nature-based infrastructure in cities is on

average 42 percent cheaper than grey infrastructure delivering the same services and generates 36 percent more value for society (Bechauf, n.d.). The cost figure comes with a condition worth stating: the 42 percent saving holds when the wider benefits of nature-based systems are included in the comparison, not when construction price is compared on its own. A ministry comparing tender prices will not see it; a ministry running a full cost-benefit appraisal may.

This shows that due to the increased value and durability of sustainable infrastructure, developing governments do not have to spend as much money on repairs, and could use those resources for other vital problems. In addition, according to Deseglise and Downing (2025), there will be a global infrastructure investment gap of approximately $15 trillion by 2040, against a total estimated investment need of roughly $97 trillion. Sustainable infrastructure can be especially useful here for developing countries, as it has a better long-term return on investment and costs less to build, which can help countries close that gap and release pressure on already strained budgets.

5. LIMITATIONS

Despite the major advantages highlighted here, sustainable infrastructure has its limitations.

Regional narrowness of the research. Some of the current research on infrastructure and developing countries may not fully represent all of them, leading to possible tunnel vision. The research cited earlier, Fay et al. (2017), is a case study focused on Latin America and the Caribbean, which does not take into account the perspectives or challenges that other countries face. Different countries face different political, social, and economic problems, so it is not possible to use a case study focused on two regions to generalize across all developing countries. However, sustainable infrastructure still has a place in most of them, because it is adaptable: with minor adjustments based on where it is being built, it can fit different environments and still provide value reliably.

Planning and governance. Another limitation is that sustainable development does not necessarily resolve all of the economic and social problems that exist in a country. According to Adam and Fazekas (2023), writing for an anti-corruption resource centre, poorly planned infrastructure can lead to corruption, inefficiency, and unequal resource distribution. This shows how important it is to have a strong government presence during the planning phases, which developing governments may struggle to maintain given the number of problems they handle at once. A possible response is for better-resourced countries or international organizations to provide planning support and oversight. It is worth noting that this limitation cuts against the cost argument specifically, since nature-based systems generally require more planning than conventional ones.

Conditional figures. The 42 percent cost advantage depends on benefit-inclusive comparison and the 20 percent outperformance figure depends on a net-zero scenario. Both are reported here with their conditions attached.

Institutional sourcing. Several figures come from organizations that advocate for the conclusions they report. They are used here because the underlying analyses are published and the direction of the evidence is consistent across sources, not because the organizations are disinterested.

6. CONCLUSION

In general, infrastructure is an essential part of development for all countries due to the basic needs it provides and its long-term implications. However, developing countries still face struggles with planning, constructing, and effectively using infrastructure. In addition, the environmental impact of conventional infrastructure is substantial. According to Goldfarb (2023), roads kill more animals than any other form of human activity, and they are set to expand, with approximately 25 million kilometres of new roads projected to be built by 2050.

Sustainable infrastructure is the best solution to these problems. It can provide economic stability through reduced lifetime costs and increased value while also maintaining the environment by interfering less with natural processes. Despite the limitations introduced here, specifically the regional narrowness of the research and the problems of planning capacity, sustainable infrastructure remains the most environmentally and economically viable option for developing countries to expand their essential infrastructure while still keeping the environment in mind.

A stronger follow-up study would compare completed nature-based and grey projects in the same country over a decade or more, measuring realized maintenance cost, service delivery, and ecological outcome rather than appraisal estimates.

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CONTRIBUTIONS

Arnav Arora, sole author, developed the research question, gathered and reviewed the sources, built the argument comparing conventional and sustainable infrastructure, and wrote the manuscript.